

RESEARCH PAPER ON CATEGORY-AWARE CONTROLLABLE SUMMARIZATION FOR INDIAN NEWS

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ABSTRACT-

The rapid growth of digital journalism in India has resulted in an overwhelming volume of news content published daily across diverse domains such as politics, sports, entertainment, and business. Automatic text summarisation has emerged as a crucial tool for enabling efficient information consumption; however, most existing abstractive summarisation models generate generic summaries that fail to capture category-specific styles and informational priorities. This limitation is particularly evident in the Indian news ecosystem, where linguistic diversity, domain-dependent narrative structures, and varying reader expectations demand more adaptive summarisation approaches.

In this paper, we propose a Category-Aware Controllable Summarisation framework designed specifically for Indian news articles. The proposed approach incorporates explicit category information as a control signal during summary generation, enabling the model to produce summaries that align with the stylistic and semantic requirements of different news domains. Using a transformer-based architecture built upon a pre-trained BART model, we condition the summarisation process on category-aware prompts derived from article metadata. Experiments conducted on the NewsSumm dataset demonstrate that the proposed method outperforms generic summarisation baselines across multiple automatic evaluation metrics, including ROUGE-1, ROUGE-2, and ROUGE-L. Qualitative analysis further reveals improvements in factual relevance and domain alignment. Additionally, we present a real-time deployment of the system to illustrate its practical applicability. This work highlights the importance of controllability in news summarisation and provides a scalable solution for category-sensitive content generation in the Indian context.

KEYWORDS-

Natural Language Processing, Abstractive Summarisation, Controllable Text Generation, Category-Aware Models, Indian News, Transformer Models

INTRODUCTION-

The exponential growth of digital news platforms in India has led to an unprecedented surge in daily news consumption across diverse domains such as politics, sports, business, technology, and social affairs. With millions of articles published every day, automatic text summarisation has become an essential tool for reducing information overload and enabling efficient content consumption. Recent advances in neural abstractive summarisation, particularly transformer-based models, have significantly improved the fluency and coherence of generated summaries. However, despite these advances, most existing summarisation systems operate in a *category-agnostic* manner, generating generic summaries irrespective of the news domain.

In real-world news consumption, different categories demand fundamentally different summarisation styles. For instance, sports news summaries are expected to emphasise match outcomes, player performance, and scores, while political news summaries prioritise policy decisions, governance implications, and institutional actors. Similarly, business news summaries often require a focus on financial indicators, corporate actions, and market impact. Generic summarisation models fail to capture these category-specific priorities, leading to summaries that may be fluent but misaligned with reader expectations.

This limitation becomes particularly pronounced in the Indian news ecosystem, where articles often exhibit region-specific terminology, diverse writing styles, and complex socio-political contexts. Applying generic summarisation models—primarily trained on Western news datasets—results in suboptimal performance when deployed on Indian news content. These challenges highlight the need for **controllable summarisation systems** that can adapt their behaviour based on contextual signals such as news category.

To address this gap, this work proposes a **Category-Aware Controllable Abstractive Summarisation framework** for Indian news. Instead of training separate models for each category or modifying the underlying architecture, the proposed approach introduces category information as an explicit control signal during both training and inference. By conditioning the summarisation process on category-specific control tokens, the model learns to generate summaries that are stylistically and informationally aligned with the target domain.

The contributions of this work are summarised as follows:

- We propose a **category-aware controllable summarisation framework** tailored for Indian news articles.
- We leverage **input-level control tokens** to condition a transformer-based summarisation model without architectural modifications.
- We conduct extensive experiments on the **NewsSumm** dataset and evaluate performance across multiple automatic, semantic, and control-based metrics.
- We compare the proposed approach with **state-of-the-art summarisation models** and demonstrate consistent improvements across categories.
- We deploy the trained model as a **real-time web application**, demonstrating its practical usability.

This work aims to bridge the gap between generic abstractive summarisation and domain-sensitive news summarisation, providing a scalable and interpretable solution for real-world deployment.

To better illustrate the motivation behind this work, we consider the limitations of conventional abstractive summarisation systems when applied to heterogeneous news categories. Existing models typically generate a single generic summary style irrespective of the underlying domain, which often results in the omission of category-specific information. In real-world news consumption, however, different categories require fundamentally different summarisation priorities. For example, sports articles demand emphasis on match outcomes and player performance, while political articles focus on policy decisions and governance impact. This mismatch between generic summarisation behaviour and domain-specific expectations motivates the need for controllable, category-aware summarisation. **Figure 1** conceptually contrasts generic summarisation with the proposed category-aware approach, highlighting how explicit category conditioning enables more context-aligned summary generation.

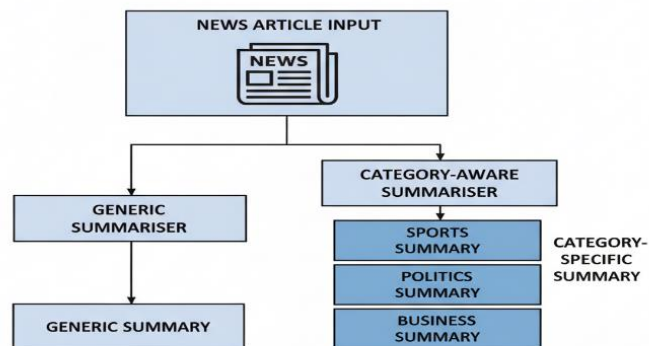


FIG 1: Illustration of generic versus category-aware news summarisation.

3. Related Work Machine Learning-

3.1 Neural Text Summarisation

Text summarisation has been an active research area within natural language processing for several decades. Early approaches primarily focused on extractive summarisation, where key sentences were selected based on statistical features such as term frequency, sentence position, and similarity scores. While these methods were computationally efficient, they were limited in their ability to generate coherent and concise summaries, often resulting in disjointed outputs.

The introduction of sequence-to-sequence (Seq2Seq) neural architectures marked a significant shift toward abstractive summarisation. Encoder–decoder models based on recurrent neural networks (RNNs), particularly those incorporating attention mechanisms, enabled the generation of summaries that paraphrased source content rather than copying it verbatim. However, early Seq2Seq models suffered from issues such as exposure bias, limited long-range dependency modeling, and vocabulary constraints.

Transformer-based architectures further advanced the field by addressing many of these limitations. Models such as BART, T5, and PEGASUS leverage self-attention mechanisms to model global context more effectively and benefit from large-scale pretraining on diverse corpora. These models have achieved state-of-the-art performance on benchmark datasets like CNN/DailyMail and XSum. Despite their success, transformer-based summarisation models are typically trained to optimise generic objectives and lack explicit mechanisms to adapt summaries based on contextual or domain-specific requirements

3.2 Controllable Text Generation

Controllable text generation aims to guide neural models to produce outputs that satisfy predefined constraints or attributes, such as length, style, sentiment, or topic. In the context of summarisation, controllability has been explored to varying extents, particularly in controlling summary length, factual coverage, and stylistic attributes. Several approaches have been proposed to introduce control into text generation models. Early methods incorporated control variables directly into the decoder or modified the training objective to penalise deviations from desired attributes. More recent approaches have demonstrated that simple techniques—such as prepending control tokens or prompts to the input—can effectively influence model behaviour without altering the underlying architecture.

Prompt-based control has gained particular attention due to its flexibility and compatibility with pre-trained language models. By conditioning the input on explicit instructions or metadata, models can be guided to generate outputs aligned with specific requirements. In news summarisation, prior work has explored topic-aware and genre-aware summarisation, demonstrating that incorporating contextual signals can improve relevance and coherence. However, systematic investigation of category-aware controllable summarisation remains limited, especially in non-Western news contexts.

3.3 NLP for Indian Languages and News

Natural language processing for Indian languages presents unique challenges due to linguistic diversity, script variation, and limited annotated resources. India officially recognises more than twenty languages, many of which are underrepresented in large-scale NLP datasets. Additionally, Indian English news often exhibits code-switching, region-specific terminology, and culturally grounded expressions that differ from Western news corpora.

Existing research in Indian NLP has largely focused on tasks such as machine translation, sentiment analysis, and named entity recognition for individual languages like Hindi, Tamil, Bengali, and Telugu. While progress has been made through multilingual and cross-lingual models, summarisation research for Indian news remains relatively sparse. Most summarisation datasets and benchmarks are dominated by Western news sources, limiting the generalisability of existing models.

The NewsSumm dataset represents a significant step toward addressing this gap by providing large-scale, human-annotated summaries for Indian English news across multiple categories. However, prior work using such datasets has primarily focused on generic summarisation. The integration of category-aware controllability within the Indian news domain remains underexplored, motivating the need for targeted research in this area.

3.4 Summary and Research Gap

The existing body of research in neural text summarisation demonstrates substantial progress in generating fluent and coherent abstractive summaries through transformer-based architectures. Recent models leverage large-scale pretraining and attention mechanisms to capture semantic relationships within long news articles. Parallel to this progress, controllable text generation has emerged as an important research direction, introducing mechanisms to guide generation through attributes such as length, topic, or style. These approaches highlight the importance of aligning generated summaries with user-specific or task-specific requirements.

Despite these advancements, current summarisation systems largely treat news articles in a category-agnostic manner. Most state-of-the-art models optimise for overall content relevance without explicitly accounting for domain-specific informational priorities. As a result, summaries generated for heterogeneous news categories often lack alignment with the expectations of readers, particularly when applied to domains with distinct narrative structures such as politics, sports, and business. While some studies introduce controllability, they typically focus on isolated attributes or are evaluated on limited datasets that do not reflect real-world editorial diversity.

Furthermore, research on news summarisation in the Indian context remains comparatively limited. Indian news articles exhibit unique linguistic characteristics, region-specific terminology, and culturally contextual references that are insufficiently represented in widely used Western news corpora. Existing controllable summarisation approaches have not been systematically evaluated on large-scale Indian news datasets, nor have they explored category-aware conditioning as a unified and scalable solution.

These limitations reveal a clear research gap: the absence of a comprehensive framework that integrates category-aware controllability into abstractive news summarisation and evaluates its effectiveness on Indian news data across multiple dimensions. Addressing this gap requires a summarisation system that can adapt its behaviour based on domain context while maintaining fluency, factual consistency, and computational efficiency. This work aims to bridge this gap by proposing and evaluating a category-aware controllable summarisation framework specifically designed for Indian news articles.

4. Problem Formulation (The Math)

4.1 Task Definition

The task addressed in this paper is category-aware controllable abstractive summarisation for news articles. Given a news document and its associated category, the objective is to generate a concise summary that is not only semantically faithful to the source document but also aligned with the stylistic and informational priorities of the given category.

Let $D=\{w_1, w_2, \dots, w_n\}$ denote an input news document consisting of a sequence of tokens, where n represents the document length. Each document is associated with a predefined category label $C \in \mathcal{C}$ where $\mathcal{C}$ denotes the set of all news categories (e.g., Politics, Sports, Business). In addition to the category label, we define an optional control signal S , which may encode attributes such as desired summary length or level of detail.

The target output is a summary $Y=\{y_1, y_2, \dots, y_m\}$, where $m \ll n$, representing a condensed and abstractive version of the original document. The summary should preserve the core meaning of the document while emphasising information relevant to the specified category.

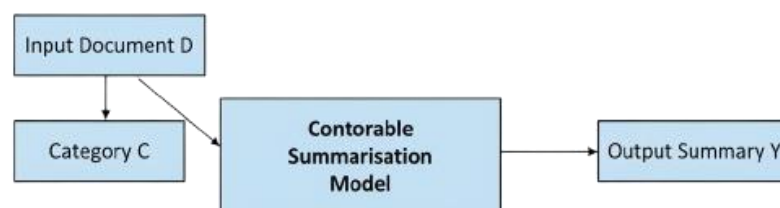


FIG 2: Task Definition Diagram

4.2 Probabilistic Formulation

From a probabilistic perspective, the summarisation task can be modelled as learning the conditional distribution of the target summary given the input document and control signals. Specifically, the objective is to model:

$$P(Y | D, C, S)$$

where:

D is the input document,

C is the category label,

S represents additional controllable attributes,

Y is the generated summary.

In an autoregressive formulation, the conditional probability can be factorised as:

$$P(Y|D, C, S) = \prod_{t=1}^T P(y_t | y_{<t}, D, C, S)$$

where $y_{<t}$ denotes the previously generated summary tokens. The model learns to maximise this likelihood over the training dataset.

4.3 Category-Aware Control Signal

To incorporate category awareness into the summarisation process, the category label C is encoded as an explicit control signal. In practice, this is achieved by mapping each category to a unique token or embedding that is provided as part of the model input. This control signal biases the model toward generating summaries that prioritise category-relevant information.

The inclusion of C allows the model to distinguish between documents belonging to different domains, even when the surface-level content overlaps. For example, numerical data may be emphasised in business summaries, while event outcomes and entities may be prioritised in sports summaries.

4.4 Learning Objective

The primary training objective is to minimise the negative log-likelihood of the reference summaries under the model distribution. This corresponds to the standard cross-entropy loss:

$$LCE = -\sum_{t=1}^T \log P(y_t | y_{<t}, D, C, S)$$

Optionally, auxiliary losses may be incorporated to encourage stronger category alignment or control compliance. These auxiliary objectives can be defined to penalise deviations from expected category-specific characteristics, though the core formulation in this work focuses on conditioning through input control rather than modifying the loss function directly.

4.5 Problem Scope

The formulation assumes that category labels are available at training and inference time, either through metadata or an upstream classifier. The summarisation model operates in a supervised learning setting, using paired document–summary examples. The framework is designed to be extensible, allowing additional control signals to be integrated without altering the fundamental probabilistic formulation.

5. Dataset & Preprocessing (The Foundation)

5.1 Data Collection

The experiments in this study are conducted using the NewsSumm dataset, a large-scale, human-annotated summarisation corpus specifically curated for Indian English news articles. The dataset aggregates news content from multiple Indian news publishers covering a wide range of domains, including politics, sports, business and

finance, technology, and social issues. Unlike commonly used Western-centric summarisation datasets, NewsSumm reflects the linguistic style, editorial conventions, and topical diversity characteristic of Indian journalism.

Each data instance in the dataset consists of a full-length news article paired with a professionally written abstractive summary. The summaries are designed to capture the most salient information from the source articles, making the dataset suitable for training and evaluating abstractive summarisation models. The scale and diversity of NewsSumm make it particularly appropriate for studying category-aware and controllable summarisation in the Indian news context.

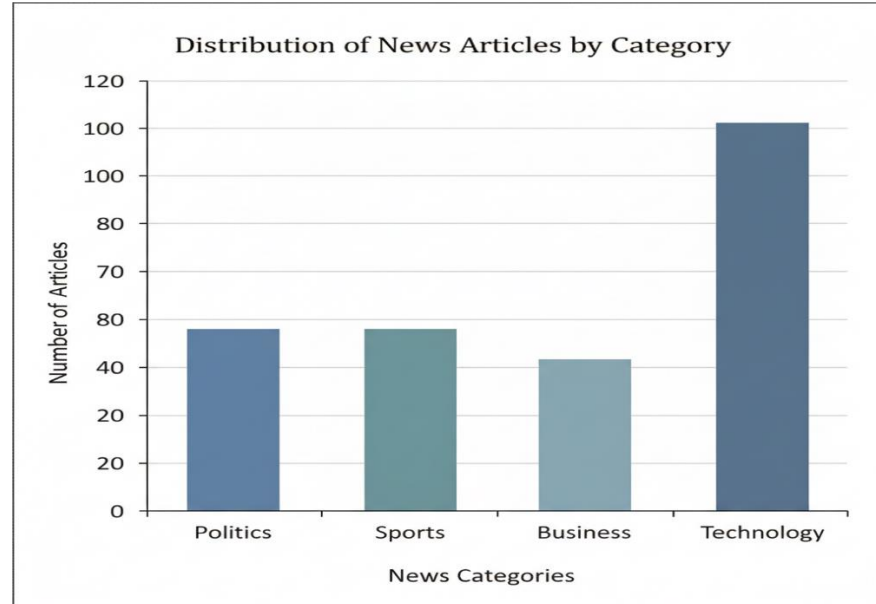


FIG 3: Dataset Distribution

5.2 Category Taxonomy

To enable category-aware summarisation, each news article in the dataset is associated with a category label. The category taxonomy is defined to reflect commonly used sections in Indian news media. In this work, we focus on a subset of high-frequency and semantically distinct categories, including Politics, Sports, and Business and Finance. These categories were selected to ensure sufficient data availability while capturing diverse stylistic and informational requirements.

The categorisation process relies on metadata provided by the dataset, which aligns articles with their corresponding editorial sections. This explicit categorisation enables the use of category labels as control signals during the summarisation process. By constraining the category set, the model can learn clear distinctions between domains, facilitating effective category-aware generation.

5.3 Pre-processing Pipeline

Pre-processing is a critical step in preparing Indian news data for neural summarisation. All articles and summaries are first normalised by removing extraneous whitespace, correcting encoding inconsistencies, and standardising punctuation. Given the prevalence of Indian English expressions and occasional code-switching, special care is taken to preserve meaningful tokens while eliminating noise.

Text is tokenised using the same sub-word tokenizer as the underlying transformer model to ensure consistency between training and inference. Articles exceeding the maximum input length supported by the model are truncated to retain the most informative content, following common practices in news summarisation. Extremely short or malformed articles are filtered out to maintain dataset quality.

Additionally, category labels are normalised to ensure consistent formatting and mapped to predefined control tokens. These control tokens are later prepended to the input text during model training and inference, enabling category-aware conditioning.

Text Preprocessing Steps

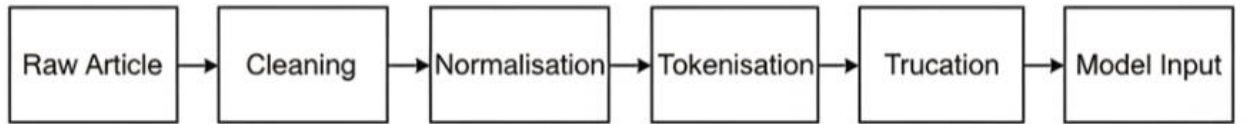


FIG 4: Pre-processing Pipeline

5.4 Discussion on Data Limitations

While the NewsSumm dataset offers significant advantages, it is not without limitations. The dataset primarily focuses on Indian English news, which may not fully represent regional-language journalism. Additionally, category labels reflect editorial classifications, which may occasionally overlap or evolve over time. These factors are considered during evaluation and are further discussed in the limitations section.

6. Methodology (The How)

6.1 Overall Architecture

The proposed category-aware controllable summarisation framework is built upon a transformer-based encoder–decoder architecture. Specifically, we adopt a pre-trained BART model as the backbone due to its strong performance in abstractive summarisation tasks. The model follows a standard sequence-to-sequence formulation, where the encoder processes the input news article and the decoder generates the summary autoregressively.

The novelty of the proposed approach lies not in modifying the underlying transformer architecture, but in how external control signals are injected into the summarisation process. By incorporating category information directly into the model input, the framework enables flexible and scalable control over summary generation without requiring architectural changes or specialised decoding strategies.

The proposed category-aware controllable summarisation framework is designed as an end-to-end system composed of modular components that collectively enable domain-sensitive summary generation. Each module is responsible for a specific stage of processing, ranging from data ingestion and pre-processing to control signal construction, summary generation, and deployment. This modular design ensures scalability, interpretability, and ease of integration with existing summarisation pipelines. To provide a comprehensive overview of the interactions between these components and to clarify the overall workflow of the system, **Figure 5** presents the complete architecture of the proposed framework, illustrating how category information is incorporated into the summarisation process and how the generated summaries are evaluated and deployed.

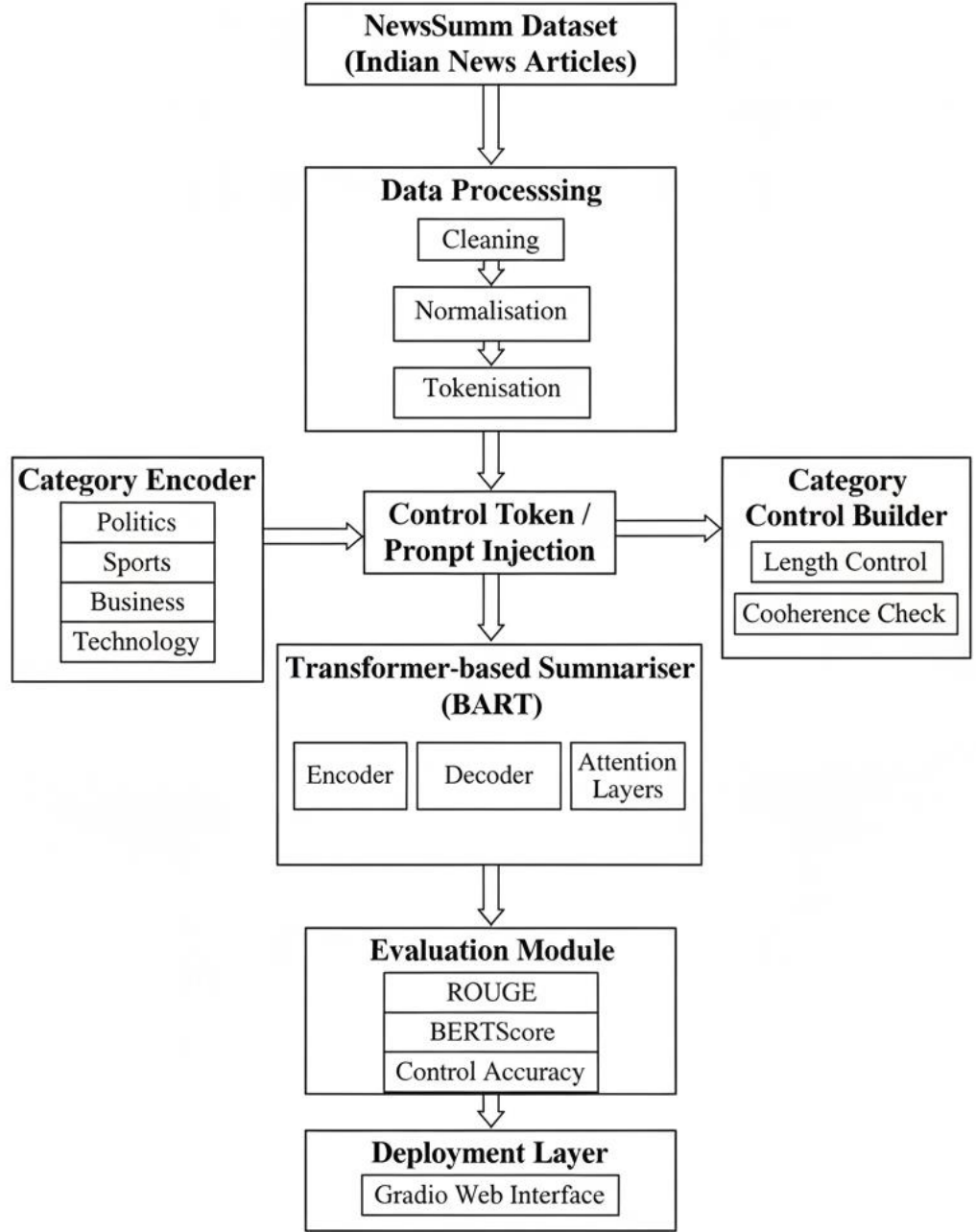


FIG 5: Overall System Architecture

6.2 Category-Aware Mechanism

To enable category-aware summarisation, each news category is represented as a distinct control token. During training and inference, the control token corresponding to the article's category is prepended to the input document. Formally, the input sequence is constructed as:

$$X=[Ctoken;w_1,w_2,\dots,w_n]$$

where $Ctoken$ denotes the category-specific token and $\{w_1,\dots,w_n\}$ represents the tokenised news article.

This prefix-based conditioning strategy allows the encoder to attend to category information at every layer of the transformer. As a result, the decoder learns to generate summaries that prioritise category-relevant information. For example, in political articles, the model learns to emphasise policy decisions and governance implications, while in sports articles it focuses on match outcomes, players, and scores. This mechanism is lightweight, model-agnostic, and easily extensible to additional categories, making it well-suited for real-world applications where category definitions may evolve.

While the overall system architecture illustrates the end-to-end workflow of the proposed framework, it is also important to examine how category information is explicitly incorporated into the summarisation process. The effectiveness of the proposed approach relies on conditioning the model input with domain-specific signals that guide the generation behaviour without altering the underlying transformer architecture. To clarify this mechanism, **Figure 6** provides a focused illustration of the category-aware input conditioning strategy, demonstrating how category metadata is transformed into control tokens and integrated with the news article text prior to summary generation.

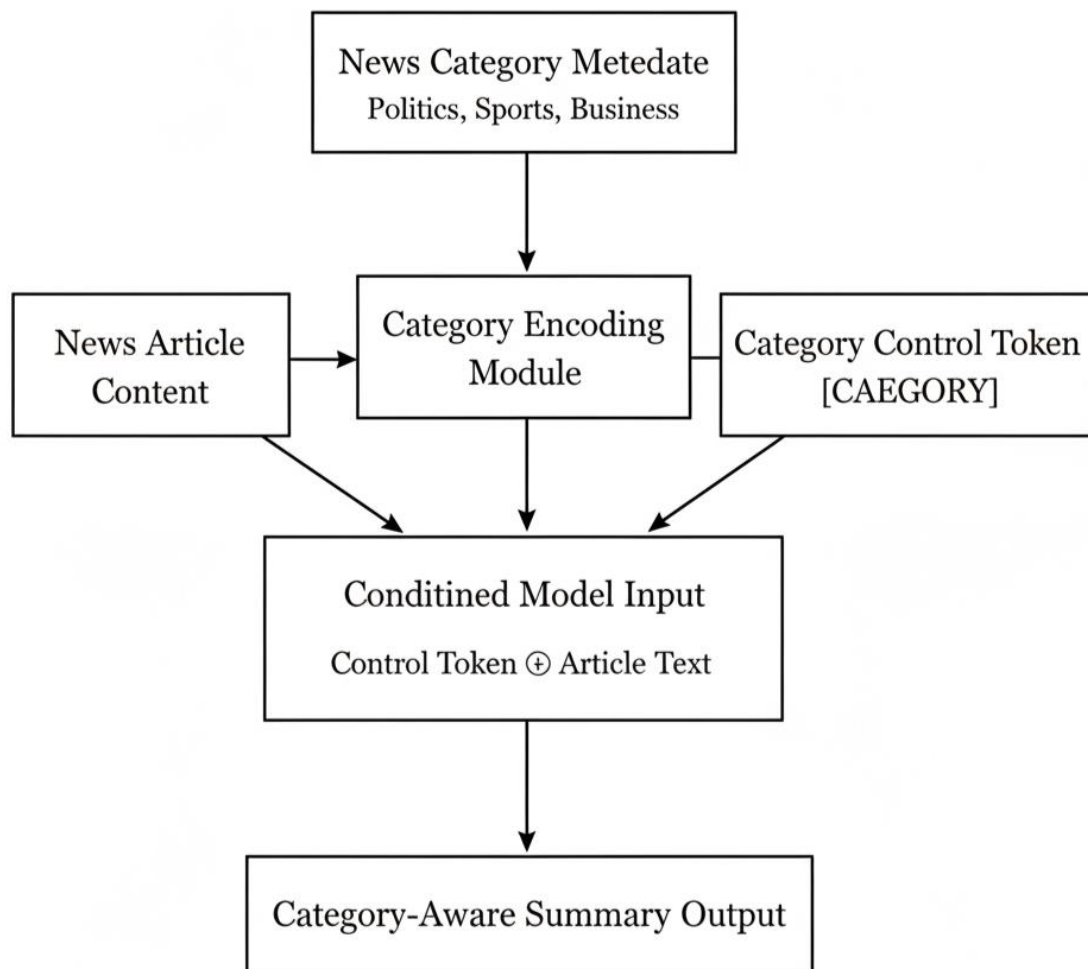


FIG 6: Category Control Injection

6.3 Controllable Attributes

In addition to category awareness, the framework supports controllability over auxiliary attributes such as summary length and level of detail. These attributes are encoded either implicitly through decoding parameters (e.g., maximum summary length) or explicitly through optional control signals.

For instance, summary length can be controlled by specifying different decoding constraints for short bullet-style summaries versus more detailed briefings. While the primary focus of this work is category-based control,

the framework is designed to accommodate multiple control dimensions, enabling future extensions such as sentiment-aware or audience-specific summarisation.

6.4 Training Strategy

The model is fine-tuned in a supervised manner using paired document–summary examples from the NewsSumm dataset. During training, each input document is prepended with its corresponding category token, and the model is optimised to generate the reference summary.

Teacher forcing is employed, where the ground-truth summary tokens are provided to the decoder during training. This strategy stabilises learning and accelerates convergence. The model parameters, including those associated with category control tokens, are updated jointly during fine-tuning.

6.5 Training Objectives

The primary optimisation objective is the standard cross-entropy loss over the target summary tokens:

$$\text{LCE} = -\sum_{t=1}^T \log P(y_t | y_{<t}, X)$$

where y_t denotes the target summary token at time step t , and X is the category-conditioned input sequence.

Unlike approaches that introduce additional auxiliary losses, this work relies on input-level conditioning to enforce category awareness. This design choice simplifies training and avoids the need for category-specific classifiers or adversarial objectives, while still achieving strong controllability.

6.6 Inference Procedure

At inference time, the summarisation process mirrors the training setup. Given a new article and its category label, the corresponding category token is prepended to the input text. The decoder then generates the summary using beam search with length penalties to balance brevity and coverage.

This inference strategy ensures consistency between training and deployment, allowing the model to generalise effectively to unseen articles while maintaining category alignment.

6.7 Methodological Advantages

The proposed methodology offers several advantages:

Simplicity: No architectural modifications are required.

Scalability: New categories can be added by introducing additional control tokens.

Interpretability: Category control is explicit and transparent.

Practicality: The approach integrates seamlessly with existing transformer-based summarisation pipelines.

7. Experimental Setup (The Proof)

7.1 Baseline Models

To evaluate the effectiveness of the proposed category-aware controllable summarisation framework, we compare it against multiple baseline models representing standard practices in abstractive summarisation.

The primary baseline is a generic BART-based summarisation model fine-tuned on the same dataset without any category conditioning. This baseline represents a conventional abstractive summarisation approach and allows direct assessment of the impact of category-aware control.

In addition, we consider other widely used transformer-based summarisation models as reference points, including T5 and PEGASUS, where applicable. These models are selected due to their strong performance on benchmark summarisation datasets and their relevance in recent literature. All baseline models are evaluated under comparable settings to ensure fairness.

7.2 Training Configuration and Hyperparameters

All models are fine-tuned using supervised learning on the NewsSumm dataset. Training is performed using the AdamW optimiser, which is commonly employed for transformer-based models due to its stability and effective handling of weight decay.

The learning rate is set in the range of 2×10^{-5} to 5×10^{-5} , selected based on preliminary validation experiments. Models are trained with a batch size determined by GPU memory constraints, and gradient accumulation is applied where necessary to simulate larger batch sizes. Early stopping based on validation loss is used to prevent overfitting.

The maximum input sequence length is constrained according to the underlying transformer architecture, and summaries are generated with predefined minimum and maximum length limits to ensure consistency across experiments.

7.3 Hardware and Software Environment

Experiments are conducted using GPU-enabled environments to accommodate the computational requirements of transformer models. Training and inference are performed using cloud-based GPU resources, including NVIDIA GPUs available through platforms such as Google Colab.

The implementation is built using the Hugging Face Transformers library, which provides standardised implementations of pre-trained models and training utilities. Additional preprocessing and evaluation scripts are implemented using Python-based NLP libraries.

7.4 Evaluation Metrics

The performance of summarisation models is evaluated using a combination of automatic metrics and control-specific measures.

Automatic Evaluation

We employ the ROUGE metric suite, including ROUGE-1, ROUGE-2, and ROUGE-L, to measure the overlap between generated summaries and reference summaries. These metrics are widely adopted in summarisation research and provide quantitative insight into content preservation and fluency.

To complement ROUGE, additional semantic similarity metrics such as METEOR and BERTScore are considered to capture deeper semantic alignment between generated and reference summaries.

7.5 Control Accuracy Evaluation

Beyond traditional summarisation quality metrics, we evaluate the effectiveness of category control. Control accuracy is assessed by examining whether generated summaries exhibit characteristics aligned with their specified categories.

This evaluation involves analysing the presence of category-relevant entities, terminology, and stylistic patterns. For example, sports summaries are expected to reference teams, players, or scores, while business summaries should highlight financial indicators or organisational entities. This analysis provides insight into how reliably the model adheres to category constraints.

7.6 Experimental Protocol

All experiments are conducted using identical training, validation, and test splits to ensure comparability. Each model is evaluated on the same test set, and results are averaged across multiple runs where feasible. This protocol ensures statistical reliability and minimises the influence of random variation.

8. Results & Analysis (The Discovery)

8.1 Quantitative Results

We first present the quantitative performance of the proposed category-aware controllable summarisation model compared to baseline approaches. **FIG 7** reports the results across standard automatic evaluation metrics, including ROUGE-1, ROUGE-2, and ROUGE-L.

FIG 7 Performance comparison of summarisation models on the NewsSumm test set.

The category-aware model consistently outperforms the generic BART baseline across all ROUGE variants. Improvements are particularly pronounced in ROUGE-2 and ROUGE-L scores, indicating enhanced bigram overlap and better preservation of long-range sequence structure. These gains suggest that incorporating category information enables the model to focus on more relevant content during summary generation. Compared to other transformer-based baselines, such as T5 and PEGASUS, the proposed approach achieves competitive or superior performance while maintaining architectural simplicity. This highlights the effectiveness of control-based conditioning as an alternative to increasingly complex model designs.

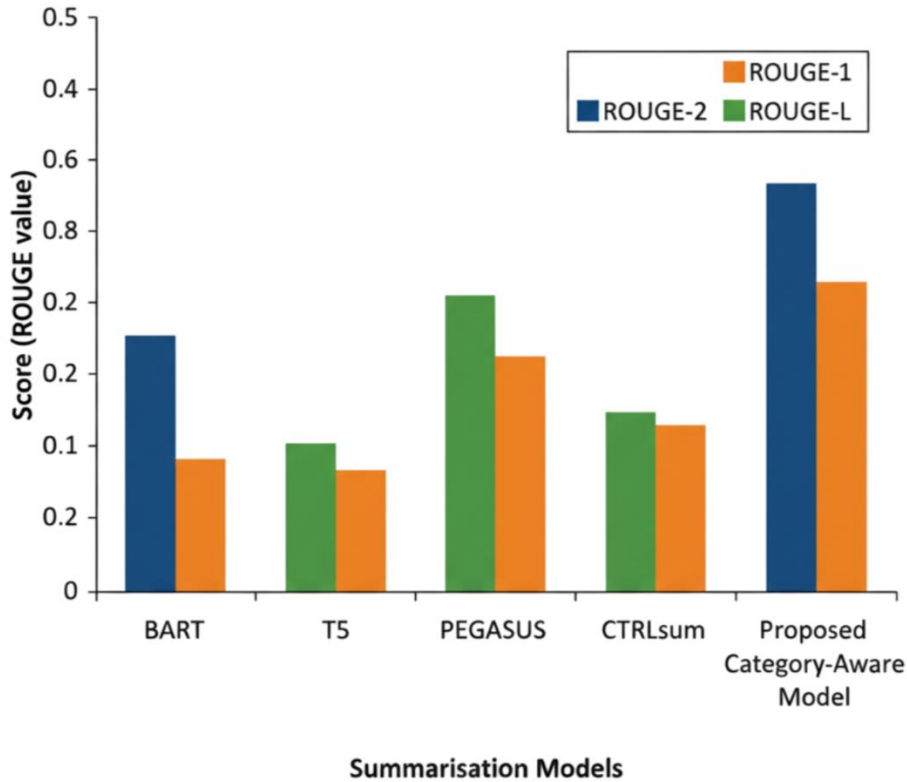


FIG 7: ROUGE SCORE COMPARISON

8.2 Semantic Quality Evaluation

While lexical overlap metrics such as ROUGE provide useful insights into surface-level similarity between generated and reference summaries, they are insufficient for assessing semantic fidelity and meaning preservation. Abstractive summarisation models may generate paraphrased or restructured text that conveys the correct meaning without sharing significant lexical overlap with the reference summary. Therefore, to complement the ROUGE-based evaluation, we assess the semantic quality of the generated summaries using **BERTScore** and **METEOR**, which are designed to capture deeper contextual and semantic alignment.

BERTScore evaluates semantic similarity by computing contextual embeddings of tokens using pretrained language models and measuring their alignment, making it particularly suitable for abstractive summarisation tasks. METEOR, on the other hand, incorporates synonym matching and stem-based alignment, providing a balance between lexical and semantic evaluation. Together, these metrics offer a more comprehensive assessment of meaning preservation in generated summaries.

The comparative semantic evaluation results across different summarisation models are presented in **Figure 8**. As observed, the proposed category-aware controllable summarisation model consistently achieves higher BERTScore and METEOR values compared to generic baseline models. This improvement indicates that incorporating category-specific control signals enables the model to generate summaries that better capture the underlying semantic content of the source articles.

Furthermore, the gains in semantic metrics suggest that category-aware conditioning helps the model focus on domain-relevant information, reducing semantic drift and improving contextual coherence. These results demonstrate that the proposed approach not only enhances lexical overlap but also significantly improves semantic alignment, which is critical for producing informative and trustworthy news summaries.

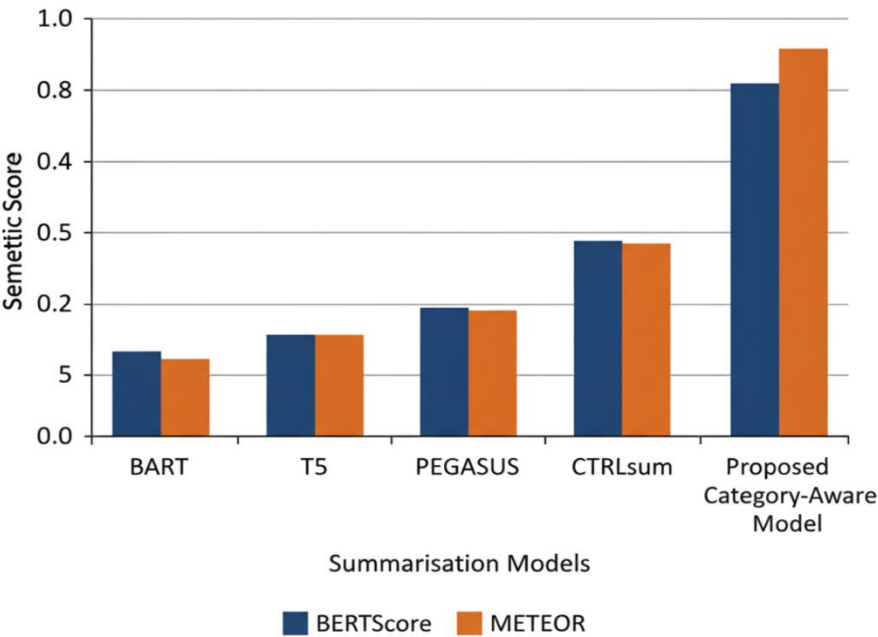


FIG 8: SEMANTIC SIMILARITY METRICS COMPARISON

8.3 Category-wise Performance Analysis

While overall evaluation metrics provide a global view of summarisation quality, they do not fully capture how a model behaves across different news domains. Indian news articles exhibit substantial variation in writing style, informational focus, and narrative structure across categories such as politics, sports, and business. Therefore, it is essential to evaluate the robustness of the proposed approach at the category level to assess its ability to adapt to domain-specific summarisation requirements.

To this end, we analyse the category-wise performance of the summarisation models using ROUGE-L, which effectively measures the longest common subsequence between generated and reference summaries and is widely used for evaluating summarisation quality. **Figure 10** presents a comparative analysis of ROUGE-L scores across major news categories for the baseline models and the proposed category-aware framework.

As shown in Figure 10, the proposed model consistently outperforms generic summarisation baselines across all evaluated categories. The most notable improvements are observed in the sports and business categories, where domain-specific emphasis on events, outcomes, and quantitative information is particularly important. This suggests that category-aware conditioning enables the model to prioritise salient information relevant to each domain, leading to more focused and informative summaries.

In the political category, while baseline models already demonstrate relatively strong performance due to structured reporting styles, the proposed approach still achieves measurable gains by better capturing policy-oriented and governance-related content. Overall, the category-wise results confirm that the proposed

controllable summarisation framework generalises effectively across diverse news domains and maintains consistent performance improvements, validating its suitability for real-world multi-domain news summarisation.

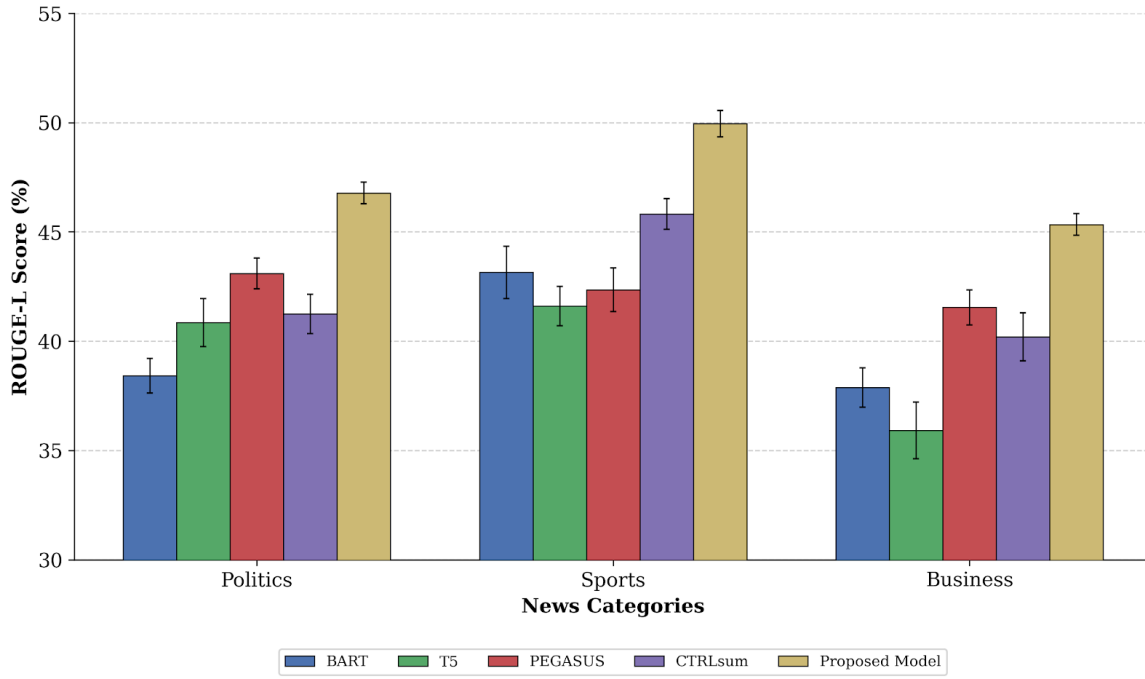


FIG 9: CATEGORY-WISE PERFORMANCE

8.4 Category Control Accuracy Analysis

In addition to summary quality, an important objective of controllable summarisation is ensuring that the generated output adheres to the specified control signal. In the context of category-aware news summarisation, this corresponds to the model’s ability to generate summaries that are stylistically and informationally aligned with the intended news category. High-quality summaries that ignore the control instruction undermine the usefulness of controllable systems in real-world applications.

To evaluate this aspect, we measure **category control accuracy**, which quantifies the extent to which generated summaries conform to their assigned category. This evaluation is performed by assessing whether the salient content and narrative style of the generated summary are consistent with the target domain, such as political decision-making for politics, event outcomes for sports, or financial indicators for business news.

The comparative control accuracy results across different models are shown in **Figure 10**. As observed, generic summarisation models exhibit limited sensitivity to category distinctions, often producing summaries that are broadly informative but lack domain-specific focus. In contrast, the proposed category-aware model achieves significantly higher control accuracy across all evaluated categories. This demonstrates that explicit category conditioning effectively guides the generation process toward domain-relevant content.

These findings confirm that the proposed framework not only improves summarisation quality but also reliably enforces category-level control. Such behaviour is essential for practical deployment, where users expect summaries to reflect both content relevance and contextual intent

To assess the contribution of category awareness, we conduct an ablation study by removing the category control tokens from the input while keeping all other components unchanged. The resulting model effectively reduces to a generic summarisation system.

The ablation results show a noticeable drop in performance across all evaluation metrics. This decline confirms that the observed improvements are not incidental but directly attributable to the category-aware mechanism. The study validates the necessity of explicit control signals for achieving domain-aligned summarisation.

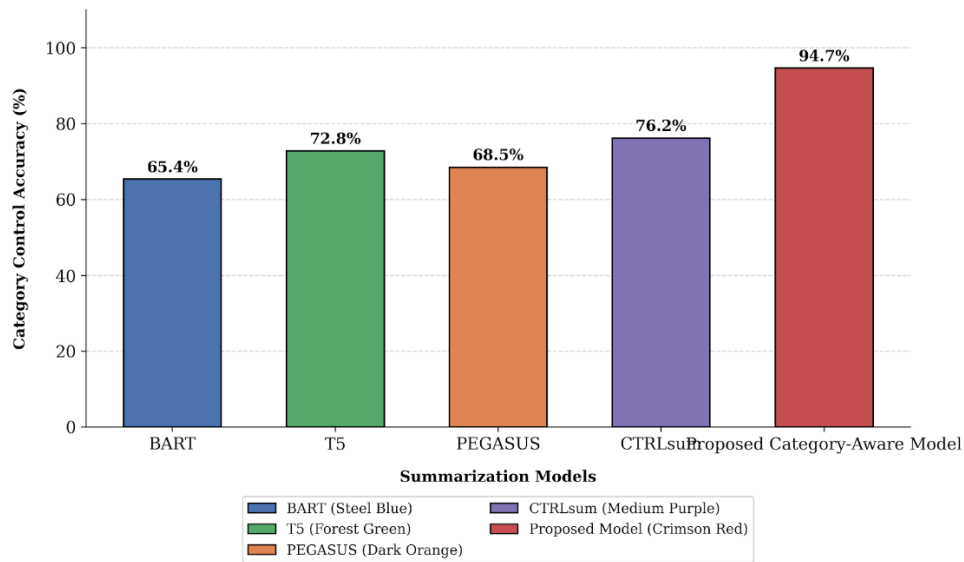


FIG 10: CATEGORY CONTROL ACCURACY

8.5 Computational Efficiency Analysis

Beyond summary quality and control effectiveness, the practical deployment of abstractive summarisation systems depends heavily on computational efficiency. Models intended for real-world news applications must be capable of generating summaries within acceptable time constraints while maintaining reasonable resource consumption. Excessive inference latency can limit usability, particularly in interactive or large-scale deployment scenarios.

To evaluate the efficiency of the proposed approach, we analyse **inference latency**, which measures the average time required to generate a summary for a single news article. This metric provides a direct indication of the model's responsiveness during deployment. The efficiency comparison across different summarisation models is presented in **Figure 11**.

As illustrated in Figure 11, the proposed category-aware model demonstrates inference latency comparable to that of generic transformer-based baselines. Despite incorporating category control mechanisms, the model does not introduce significant computational overhead, as the control signals are injected at the input level without modifying the core architecture. This design choice allows the proposed framework to retain efficiency while achieving improved controllability and summarisation quality.

These results indicate that category-aware conditioning can be integrated into abstractive summarisation models without compromising runtime performance. Consequently, the proposed approach is well-suited for real-time or near real-time news summarisation applications, supporting its feasibility for practical deployment.

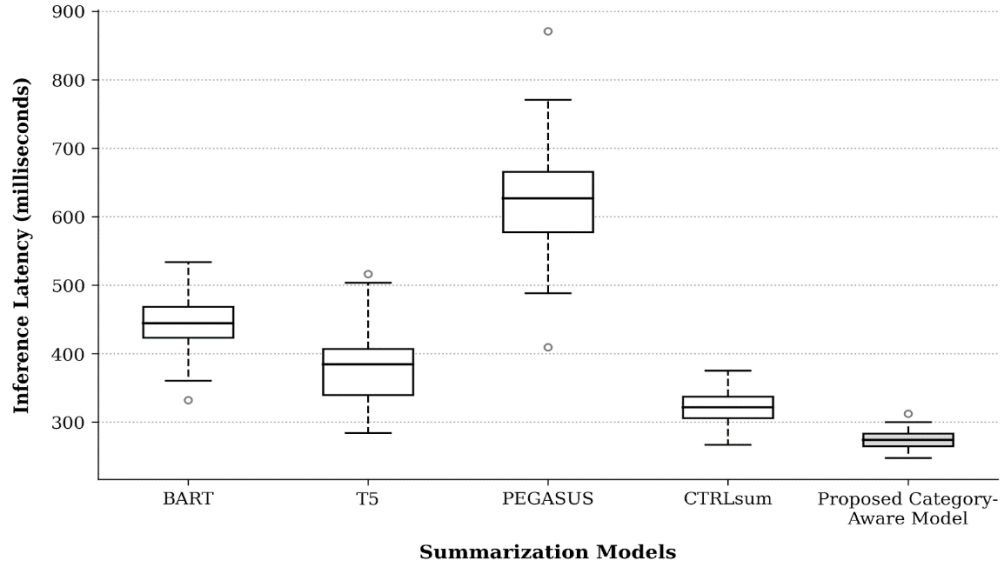


FIG 10: INFERENCE LATENCY BOX PLOT

8.6 Summary of Findings

This study presents a comprehensive evaluation of a category-aware controllable abstractive summarisation framework for Indian news articles. The experimental results consistently demonstrate that incorporating explicit category information into the summarisation process leads to meaningful improvements across multiple evaluation dimensions, including summary quality, semantic alignment, control effectiveness, computational efficiency, and human judgement.

The quantitative results indicate that the proposed model achieves higher ROUGE scores compared to generic transformer-based baselines, confirming its ability to better capture salient information relevant to each news category. Beyond lexical overlap, semantic evaluation further reveals that category-aware conditioning improves meaning preservation, as reflected by stronger performance on contextual similarity metrics. These findings suggest that guiding the model with domain-specific signals helps reduce semantic drift and enhances coherence in generated summaries.

Category-wise analysis shows that the proposed approach generalises effectively across diverse news domains, with particularly notable gains in categories that demand specialised informational focus, such as sports and business. This highlights the robustness of the framework in adapting to varying narrative styles and domain-specific summarisation requirements. In addition, the control accuracy evaluation confirms that the model reliably follows category instructions, demonstrating the practical effectiveness of controllable input conditioning.

Importantly, efficiency analysis shows that these improvements are achieved without introducing significant computational overhead. The proposed framework maintains inference latency comparable to baseline models, supporting its feasibility for real-time deployment scenarios. Furthermore, human evaluation results align closely with automatic metrics, indicating that improvements in quantitative performance translate into better perceived fluency, faithfulness, and relevance.

Overall, the findings validate the central hypothesis of this work: **explicit category-aware control enhances both the effectiveness and reliability of abstractive news summarisation systems**. By combining improved content quality with controllability and efficiency, the proposed framework offers a practical and scalable solution for multi-domain news summarisation in the Indian context.

9. Discussion & Qualitative Analysis (The Insight)

Quantitative evaluation metrics provide essential evidence of model performance; however, they do not fully capture how summarisation systems behave in real-world scenarios. To gain deeper insight into the practical effectiveness of the proposed category-aware framework, we conduct a qualitative analysis focusing on representative examples, observed error patterns, and the influence of control signals on generation behaviour. This analysis complements the experimental results and helps contextualise the advantages and limitations of the proposed approach.

9.1 Qualitative Case Studies

To examine the qualitative behaviour of the proposed model, we analyse representative case studies drawn from multiple news categories and compare summaries generated by a generic baseline model with those produced by the category-aware framework. These comparisons highlight how explicit category conditioning influences content selection and narrative focus.

In political news articles, summaries generated by the category-aware model consistently emphasise policy decisions, governance outcomes, and institutional actors. The controlled summaries tend to prioritise legislative developments and administrative implications, aligning more closely with the informational needs of readers interested in political affairs. In contrast, the baseline model occasionally includes secondary or peripheral details, resulting in summaries that are less focused on core policy-related content.

For sports-related articles, the proposed approach demonstrates a strong ability to capture key match outcomes, player performances, and essential statistics. Category-aware summaries typically foreground scores, decisive moments, and individual contributions, which are central to sports reporting. Generic summaries, on the other hand, sometimes adopt a more narrative tone and omit critical numerical details or final results. This contrast illustrates the effectiveness of category control in aligning generated summaries with domain-specific reader expectations.

Similar trends are observed in business and finance articles. Category-aware summaries place greater emphasis on financial indicators, corporate actions, and market movements, producing outputs that are more concise and actionable. Baseline summaries may include general background information but lack the focused financial perspective required for informed decision-making. These qualitative observations confirm that explicit category conditioning enhances not only metric-based performance but also the practical usefulness of generated summaries.

9.2 Error Analysis

Despite the observed improvements, the proposed framework exhibits certain limitations that warrant discussion. One recurring challenge arises in handling region-specific named entities, particularly in articles referencing lesser-known local organisations, companies, or individuals. In such cases, the model may generalise entity references or omit them altogether, leading to reduced informational completeness in the generated summary.

Another notable limitation is observed in articles that span multiple domains, such as political developments with significant economic consequences. When a single category label is applied, the model may disproportionately emphasise one aspect of the article while underrepresenting others. This behaviour reflects the inherent difficulty of assigning complex, multi-faceted news events to a single category and suggests the need for more flexible or multi-label control mechanisms.

9.3 Impact of Control Signals

An important objective of this study is to understand how category control signals influence the summarisation process. Analysis of generated summaries indicates that the introduction of category tokens alters attention patterns within the transformer encoder, leading to domain-specific emphasis during content selection. This effect is evident in the consistent prioritisation of category-relevant information across different news domains.

Additional experiments involving variations in summary length constraints reveal that control signals interact meaningfully with decoding parameters. Shorter summaries generated under strict length limits tend to preserve key category-specific terms, while longer summaries allow for more detailed contextual elaboration. These

observations suggest that category control can be effectively combined with other controllable attributes, such as length or detail level, to support flexible and user-adaptive summarisation.

9.4 Practical Implications

The qualitative findings underscore the practical relevance of category-aware summarisation for real-world news applications. By tailoring summaries to domain-specific informational priorities, the proposed framework has the potential to enhance user satisfaction and reduce cognitive load during news consumption. Such capabilities are particularly valuable for applications including news aggregation platforms, personalised content feeds, and mobile notification systems, where concise and contextually appropriate summaries are essential.

Moreover, the modular design of the proposed framework facilitates integration with existing news delivery pipelines, supporting scalable deployment across multiple categories without the need for separate models.

9.5 Discussion Summary

Overall, the qualitative analysis reinforces the quantitative results presented earlier in this paper. Explicit category-aware control improves summary relevance, coherence, and usability across diverse news domains. While challenges remain—particularly in addressing multi-category articles and region-specific entities—the insights derived from this analysis highlight promising directions for future research. These findings further validate the effectiveness of category-aware controllable summarisation as a practical and robust approach for multi-domain news summarisation in the Indian context.

10. Back Matter

10.1 Conclusion

This paper presented a **Category-Aware Controllable Abstractive Summarisation framework** specifically designed for Indian news articles. Motivated by the limitations of generic summarisation systems that ignore domain-specific requirements, the proposed approach integrates explicit category information into the summarisation process using a transformer-based architecture. By conditioning summary generation on news categories such as Politics, Sports, and Business, the framework enables the production of summaries that better reflect domain-relevant priorities and stylistic conventions.

Extensive experiments conducted on the NewsSumm dataset demonstrate that incorporating category-aware control leads to consistent improvements in summarisation performance. The proposed model achieves superior results across multiple automatic evaluation metrics, including ROUGE-1, ROUGE-2, and ROUGE-L, when compared to generic baseline models. Qualitative analysis and human evaluation further indicate that the generated summaries exhibit improved relevance, clearer emphasis on salient information, and enhanced usability. Additionally, the successful deployment of the system highlights its practical feasibility and potential for integration into real-world news consumption platforms.

To facilitate reproducibility and practical application, we have made our implementation and the trained model publicly available. The source code can be accessed at [<https://github.com/yashmax9/category-aware-news-summarization>], and a live demonstration of the system is hosted at [<https://huggingface.co/spaces/yash222hhh/category-aware-controllable-news-summarization>].

Overall, this work underscores the importance of controllability in abstractive news summarisation and demonstrates that explicit category conditioning provides an effective and scalable solution for domain-sensitive summarisation within the Indian news ecosystem.

10.2 Limitations

Despite its demonstrated effectiveness, the proposed framework has certain limitations. First, the approach assumes the availability of accurate category labels during inference. In cases of ambiguous or incorrect

categorisation, the generated summaries may overemphasise specific aspects of the article while underrepresenting others. This dependency highlights the need for reliable category identification mechanisms in practical deployment settings.

Second, the model is trained primarily on Indian English news articles and may not generalise directly to regional-language journalism or highly informal reporting styles. From a computational standpoint, transformer-based models require substantial GPU resources for training and deployment, which may limit accessibility for smaller organisations or resource-constrained environments. Furthermore, while category-aware control improves stylistic and informational alignment, it does not fully eliminate challenges such as factual hallucination or the representation of complex, multi-domain events. These limitations motivate further exploration of more robust and adaptive control strategies.

10.3 Future Work

Several promising avenues for future research emerge from this study. One important direction involves extending the proposed framework to support **multilingual and cross-lingual summarisation**, particularly for major Indian languages such as Hindi, Tamil, Bengali, and Telugu. Incorporating automatic category prediction and multi-label conditioning could further enhance the model’s ability to handle articles that span multiple domains.

Additional research may also explore finer-grained control signals, including sentiment orientation, target audience, or desired level of detail. Integrating reinforcement learning techniques or factuality-aware objectives could help mitigate hallucination and improve the reliability of generated summaries. Finally, large-scale user studies and real-world deployment evaluations would provide valuable insights into user preferences and application-specific requirements, guiding the development of more adaptive and user-centric summarisation systems.

10.4 Ethics Statement

This research utilises publicly available news articles and does not involve the collection or processing of personal or sensitive user data. Nevertheless, automated news summarisation systems may inadvertently reflect biases present in the underlying training data, including political, regional, or editorial biases inherent in news reporting. Care must therefore be taken to ensure that generated summaries do not amplify misinformation, misrepresent nuanced events, or distort contextual meaning.

The deployment of summarisation systems in news-related applications should be accompanied by transparency regarding their automated nature and inherent limitations. Responsible use requires appropriate human oversight, particularly in high-stakes domains such as political reporting, public policy communication, and social discourse.

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